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United States Patent	12413185
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Wu; Yan-Zheng et al.

Electronic device

Abstract

An electronic device, including a substrate and multiple modulation units, is provided. The modulation units are disposed on the substrate. Each modulation unit includes a first electronic element, a second electronic element, a first signal line, a second signal line, and a third signal line. The first signal line provides a first voltage to the first electronic element. The second signal line provides a second voltage to the second electronic element. The third signal line provides a third voltage to the first electronic element and/or the second electronic element. The first voltage is different from the second voltage, and the third voltage is different from the first voltage and/or the second voltage.

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Appl. No.: **17/955510**

Filed: **September 28, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230118076 A1	Apr. 20, 2023

Foreign Application Priority Data

CN	202111223320.X	Oct. 20, 2021

Publication Classification

Int. Cl.: **H03C7/02** (20060101); **H01P3/12** (20060101); **H03C3/00** (20060101); **H03H7/00** (20060101); **H04B1/04** (20060101)

U.S. Cl.:

CPC **H03C7/02** (20130101); **H01P3/12** (20130101); **H03C3/00** (20130101); **H03H7/00** (20130101); **H04B1/04** (20130101);

Field of Classification Search

CPC: H03C (7/02); H03C (3/00); H01P (3/12); H03H (7/00); H04B (1/04)

USPC: 332/185

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims the priority benefit of China application serial no. 202111223320.X, filed on Oct. 20, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.
BACKGROUND
Technical Field
(2) The disclosure relates to an electronic device, and in particular to an electronic device having a

modulation unit capable of modulating a phase, a bandwidth, an intensity, or a polarization state of an electromagnetic wave.

Description of Related Art

(3) Electronic devices or splicing electronic devices have been widely used in mobile phones, televisions, monitors, tablet computers, car displays, wearable devices, and desktop computers. With the vigorous development of electronic devices, the quality requirements for electronic devices are increasing, and such electronic products can often be used as electronic modulation devices at the same time, for example, as antenna devices that can modulate electromagnetic waves. However, the existing antenna devices still do not meet the requirements of consumers in all aspects.

SUMMARY

(4) The disclosure provides an electronic device having a modulation unit capable of modulating a phase, a bandwidth, an intensity, or a polarization state of an electromagnetic wave.

(5) According to an embodiment of the disclosure, the electronic device includes a substrate and multiple modulation units. The modulation units are disposed on the substrate. Each modulation unit includes a first electronic element, a second electronic element, a first signal line, a second signal line, and a third signal line. The first signal line provides a first voltage to the first electronic element. The second signal line provides a second voltage to the second electronic element. The third signal line provides a third voltage to the first electronic element and/or the second electronic element. The first voltage is different from the second voltage, and the third voltage is different from the first voltage and/or the second voltage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings are included to provide a further understanding of the disclosure, and the drawings are incorporated into the specification and constitute a part of the specification. The drawings illustrate embodiments of the disclosure and serve to explain principles of the disclosure together with the description.

(2) FIG. **1**A is a schematic partial top view of an electronic device according to an embodiment of the disclosure.

(3) FIG. **1**B is a schematic cross-sectional view of the electronic device of FIG. **1**A along a section line I-I′.

(4) FIG. **2**A is a schematic partial top view of an electronic device according to another embodiment of the disclosure.

(5) FIG. **2**B is a schematic cross-sectional view of the electronic device of FIG. **2**A along a section line II-II′.

(6) FIG. **3**A is a schematic partial top view of an electronic device according to another embodiment of the disclosure.

(7) FIG. **3**B is a schematic cross-sectional view of the electronic device of FIG. **3**A along a section line III-III′.

(8) FIG. **4**A is a schematic partial top view of an electronic device according to another embodiment of the disclosure.

(9) FIG. **4**B is a schematic cross-sectional view of the electronic device of FIG. **4**A along a section line IV-IV′.

(10) FIG. **5** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.

(11) FIG. **6** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.

(12) FIG. **7** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(13) FIG. **8** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(14) FIG. **9** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(15) FIG. **10** is a functional schematic view of an electronic device according to another embodiment of the disclosure.
(16) FIG. **11** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(17) FIG. **12** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(18) FIG. **13** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(19) FIG. **14** is a schematic partial top view of an electronic device according to another embodiment of the disclosure.
(20) FIG. **15** to FIG. **17** are schematic partial top views of an electronic device according to multiple embodiments of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

(21) The disclosure may be understood by referring to the following detailed description in conjunction with the drawings. It should be noted that in order to facilitate understanding by the reader and the simplicity of the drawings, multiple drawings in the disclosure only depict a part of an electronic device, and specific elements in the drawings are not drawn according to actual scale. In addition, the number and the size of each element in the drawings are for illustration only and are not intended to limit the scope of the disclosure.
(22) In the following description and claims, words such as “comprising” and “including” are open-ended words, so the words should be interpreted as meaning “comprising but not limited to . . . ”.
(23) It will be understood that when an element or a film layer is referred to as being “on” or “connected to” another element or film layer, the element or the film layer may be directly on the other element or film layer or directly connected to the other element or film layer, or there may be an intervening element or film layer between the two (indirect case). In contrast, when an element is referred to as being “directly on” or “directly connected to” another element or film layer, there is no intervening element or film layer between the two.
(24) Although the terms “first”, “second”, “third” . . . may be used to describe various constituent elements, the constituent elements are not limited by the terms. The terms are only used to distinguish a single constituent element from other constituent elements in the specification. The same terms may not be used in the claims, but replaced by first, second, third . . . according to the order in which the elements are declared in the claims. Therefore, a first constituent element in the following specification may be a second constituent element in the claims.
(25) In the text, the terms “about”, “approximately”, “substantially”, and “roughly” generally mean within 0.5%, 1%, 2%, 3%, 5%, or 10% of a given value or range. The number given here is an approximate number, that is, the meaning of “about”, “approximately”, “substantially”, or “roughly” may still be implied without the specific description of “about”, “approximately”, “substantially”, and “roughly”.
(26) In some embodiments of the disclosure, terms related to bonding and connection such as “connection” and “interconnection” may mean that two structures are in direct contact or may also mean that two structures are not in direct contact, wherein there is another structure disposed between the two structures, unless otherwise defined. Also, the terms related to bonding and connection may also include the case where the two structures are both movable or the two

structures are both fixed. Furthermore, the term "coupling" includes any direct and indirect means of electrical connection.
(27) The electronic device may include a display device, an antenna device (for example, a liquid crystal antenna), a sensing device, a light emitting device, a touch device, or a splicing device, but not limited thereto. The electronic device may include a bendable or flexible electronic device. The appearance of the electronic device may be rectangular, circular, polygonal, a shape with curved edges, or other suitable shapes. The display device may, for example, include a light emitting diode (LED), liquid crystal, fluorescence, phosphor, quantum dot (QD), other suitable materials, or a combination of the foregoing, but not limited thereto. The light emitting diode may include, for example, an organic light emitting diode (OLED), an inorganic light emitting diode, a mini LED, a micro LED, a QDLED, other suitable materials, or any combination of the foregoing, but not limited thereto. The display device may also include, for example, a splicing display device, but not limited thereto. The antenna device may be, for example, a liquid crystal antenna, but not limited thereto. The antenna device may include, for example, an antenna splicing device, but not limited thereto. It should be noted that the electronic device may be any combination of the foregoing, but not limited thereto. In addition, the appearance of the electronic device may be rectangular, circular, polygonal, a shape with curved edges, or other suitable shapes. The electronic device may have a peripheral system such as a driving system, a control system, a light source system, and a shelf system to support the display device, the antenna device, or the splicing device. The disclosure will be described below with the electronic device, but the disclosure is not limited thereto.
(28) It should be noted that in the following embodiments, features in several different embodiments may be replaced, recombined, and mixed to complete other embodiments without departing from the spirit of the disclosure. As long as the features between various embodiments do not violate the spirit of the invention or conflict with each other, the features may be mixed and matched arbitrarily.
(29) Reference will now be made in detail to the exemplary embodiments of the disclosure, examples of which are illustrated in the drawings. Wherever possible, the same reference numerals are used in the drawings and the description to refer to the same or similar parts.
(30) FIG. **1**A is a schematic partial top view of an electronic device according to an embodiment of the disclosure. FIG. **1**B is a schematic cross-sectional view of the electronic device of FIG. **1**A along a section line I-I′. For the clarity of the drawings and the convenience of description, FIG. **1**A omits to show several elements in an electronic device **10**.
(31) Please refer to FIG. **1**A and FIG. **1**B at the same time. The electronic device **10** of this embodiment includes a substrate **101** and multiple modulation units **100**. The modulation units **100** are disposed on the substrate **101** and are used to modulate an intensity, a bandwidth, or a phase of a received electromagnetic wave signal (or light signal, but not limited thereto). Specifically, each modulation unit **100** includes a first conductive layer pattern **120**, a first insulation layer **130**, a first electronic element **141**, a second electronic element **142**, a first signal line **151**, a second signal line **152**, and a third signal line **153**. The substrate **101** may include a hard substrate, a soft substrate, or a combination of the foregoing. For example, the material of the substrate **101** may include glass, quartz, silicon wafer, sapphire, III-V semiconductor material, ceramics, polycarbonate (PC), polyimide (PI), polyethylene terephthalate (PET), other suitable substrate materials, or a combination of the foregoing, but not limited thereto. In some embodiments, the substrate **101** may be a printed circuit board.
(32) In this embodiment, a conductive layer including the first conductive layer pattern **120** is disposed on the substrate **101**. The first conductive layer pattern **120** includes a first part **121**, a second part **122**, and a third part **123**. The first part **121**, the second part **122**, and the third part **123** are separated from each other. The third part **123** is located between the first part **121** and the second part **122**. In this embodiment, the first conductive layer pattern **120** may resonate with the

received electromagnetic wave signal (or light signal). The material of the first conductive layer pattern **120** may include copper, aluminum, silver, gold, indium tin oxide (ITO), metal alloy, other suitable conductive materials, or a combination of the foregoing materials, but not limited thereto.

(33) In this embodiment, in the schematic top view of the modulation unit **100** (as shown in FIG. **1**A), the pattern of the first conductive layer pattern **120** may include the pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123**. The pattern of the first part **121** and the pattern of the second part **122** may be regarded as E-shaped, and the pattern of the third part **123** may be regarded as a solid square shape, but not limited thereto. The pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** are separated from each other. The pattern of the first part **121** and the pattern of the second part **122** may surround the pattern of the third part **123**, and the pattern of the first part **121** and the pattern of the second part **122** may be disposed on two sides of the pattern of the third part **123** in a mirroring manner, but not limited thereto.

(34) In this embodiment, the first part of the insulation layer, that is, the first insulation layer **130** is disposed on the first conductive layer pattern **120** to cover a part of the first conductive layer pattern **120** and cover a part of the substrate **101** exposed by an opening between the first conductive layer patterns **120**. The first insulation layer **130** may expose another part of the first conductive layer pattern **120** for disposing a bonding pad **161**, a bonding pad **162**, a bonding pad **163**, and a bonding pad **164**. In this embodiment, the first insulation layer **130** may be a single-layer or multi-layer structure, and the material of the first insulation layer **130** may include a polymer thin film, silicon nitride, silicon oxide, silicon oxynitride, or a combination of the foregoing, but not limited thereto.

(35) In this embodiment, each modulation unit **100** further includes a bonding pad **161**, a bonding pad **162**, a bonding pad **163**, and a bonding pad **164**, wherein the bonding pad **161** and the bonding pad **163** are respectively disposed between the first conductive layer pattern **120** and the first electronic element **141**, and the bonding pad **162** and the bonding pad **164** are respectively disposed between the first conductive layer pattern **120** and the second electronic element **142**. The bonding pad **161** is disposed on the first part **121** of the first conductive layer pattern **120** to contact the first part **121**. The bonding pad **162** is disposed on the second part **122** of the first conductive layer pattern **120** to contact the second part **122**. The bonding pad **163** and the bonding pad **164** are respectively disposed on the third part **123** of the first conductive layer pattern **120** to contact the third part **123**. In this embodiment, the materials of the bonding pad **161**, the bonding pad **162**, the bonding pad **163**, and the bonding pad **164** may include copper, aluminum, silver, gold, indium tin oxide, metal alloys (for example, electroless nickel immersion gold (ENIG)), other suitable conductive materials, or a combination of the foregoing materials, but not limited thereto.

(36) In this embodiment, each modulation unit **100** further includes multiple solder balls **165**, **166**, **167**, and **168**, which are respectively disposed on the bonding pad **161**, the bonding pad **162**, the bonding pad **163**, and the bonding pad **164**. The solder ball **165** may contact the bonding pad **161**, the solder ball **166** may contact the bonding pad **162**, the solder ball **167** may contact the bonding pad **163**, and the solder ball **168** may contact the bonding pad **164**.

(37) In this embodiment, the first electronic element **141** and the second electronic element **142** are disposed on the first signal line **151**, the second signal line **152**, and the third signal line **153**. The first electronic element **141** has a pad **141***a* and a pad **141***b*, and the second electronic element **142** has a pad **142***a* and a pad **142***b*. The first electronic element **141** may be bonded to the bonding pad **161** through the pad **141***a* and the solder ball **165**, and may be bonded to the bonding pad **163** through the pad **141***b* and the solder ball **167**. The second electronic element **142** may be bonded to the bonding pad **162** through the pad **142***a* and the solder ball **166**, and may be bonded to the bonding pad **164** through the pad **142***b* and the solder ball **168**. In other words, the first electronic element **141** may be bonded to the first conductive layer pattern **120** through the bonding pad **161** and the bonding pad **163**, and the second electronic element **142** may be bonded to the first

conductive layer pattern **120** through the bonding pad **162** and the bonding pad **164**.

(38) In this embodiment, the first electronic element **141** and the second electronic element **142** may have the characteristic of being sensitive to external stimuli, and the first electronic element **141** and the second electronic element **142** may be regulated by the external stimuli to change their own characteristics, wherein the external stimulus is, for example, voltage, current, temperature, etc., and their own characteristic is, for example, a capacitance value or a resistance value, but not limited thereto. In this embodiment, the first electronic element **141** and the second electronic element **142** may include, for example, capacitors, inductors, or resistors, but not limited thereto. For example, when the first electronic element **141** and/or the second electronic element **142** are capacitors, the modulation unit **100** may regulate the capacitors by applying voltages to change the capacitance values of the capacitors.

(39) In addition, in some embodiments, the first electronic element **141** and the second electronic element **142** may also be, for example, electronic elements containing materials sensitive to external stimuli, electronic elements containing elements sensitive to external stimuli, or electronic elements containing structures sensitive to external stimuli. The material sensitive to external stimuli may be, for example, liquid crystal, vanadium oxide, etc., the element sensitive to external stimuli may be, for example, a PIN diode, a varactor diode, a Schottky diode, a Gunn diode, etc., the structure sensitive to external stimuli may be, for example, micro electro mechanical systems (MEMS), etc., but not limited thereto.

(40) In addition, in some embodiments, the first electronic element **141** and the second electronic element **142** may also be different electronic elements. For example, the first electronic element **141** may be an inductor and the second electronic element **142** may be a capacitor or the first electronic element **141** may be a switching element and the second electronic element **142** may be a diode. Therefore, one modulation element may modulate two different characteristics.

(41) In this embodiment, the first signal line **151**, the second signal line **152**, and the third signal line **153** are respectively disposed on the substrate **101** and the first insulation layer **130**. The first signal line **151**, the second signal line **152**, and the third signal line **153** may be the same film layer as the first conductive layer pattern **120**, but not limited thereto. The first signal line **151** may contact and be coupled to the first part **121** of the first conductive layer pattern **120**, the second signal line **152** may contact and be coupled to the second part **122** of the first conductive layer pattern **120**, and the third signal line **153** may contact and be coupled to the third part **123** of the first conductive layer pattern **120**.

(42) In this embodiment, the first signal line **151** has a first voltage V**1**, the second signal line **152** has a second voltage V**2**, and the third signal line **153** has a third voltage V**3**. The first voltage V**1**, the second voltage V**2**, and the third voltage V**3** are low frequency voltage signals, for example, the frequency range is 0 to 100 MHz. The first signal line **151** may provide the first voltage V**1** to the first electronic element **141** through the first part **121**, the bonding pad **161**, the solder ball **165**, and the pad **141***a*. The second signal line **152** may provide the second voltage V**2** to the second electronic element **142** through the second part **122**, the bonding pad **162**, the solder ball **165**, and the pad **142***b*. The third signal line **153** may provide the third voltage V**3** to the first electronic element **141** through the third part **123**, the bonding pad **163**, the solder ball **165**, and the pad **141***b*, and provide the third voltage V**3** to the second electronic element **142** through the third part **123**, the bonding pad **164**, the solder ball **165**, and the pad **142***a*. Therefore, in this embodiment, the first electronic element **141** may be regulated and the own characteristics of the first electronic element **141** may be changed using the third voltage V**3** of the third signal line **153** matched with the first voltage V**1** of the first signal line **151** or the second electronic element **142** may be regulated and the own characteristics of the second electronic element **142** may be changed using the third voltage V**3** of the third signal line **153** matched with the second voltage V**2** of the second signal line **152**. In this way, the modulation unit **100** may modulate the phase, the intensity, the bandwidth, or the polarization state of the received electromagnetic wave signal (or light signal, but

not limited thereto), and output the modulated electromagnetic wave signal (or light signal, but not limited thereto).
(43) In addition, in this embodiment, a voltage difference (that is, a difference value between the first voltage V**1** and the third voltage V**3**) in the first electronic element **141** and a voltage difference (that is, a difference value between the first voltage V**1** and the second voltage V**2**) in the second electronic element **142** may be independently regulated by the first signal line **151**, the second signal line **152**, and/or the third signal line **153**. In this embodiment, since the first electronic element **141** and the second electronic element **142** may share the third voltage V**3** of the third signal line **153**, the overall circuit configuration of the modulation unit **100** can be relatively simple and not complicated. In this embodiment, since the first voltage V**1** may be different from the second voltage V**2**, and the third voltage V**3** may be different from the first voltage V**1** and the second voltage V**2**, the regulated first electronic element **141** and second electronic element **142** may respectively display various different characteristics, thereby increasing the modulable factor of the modulation unit **100** or increasing the selection of the first electronic element **141** and the second electronic element **142**.
(44) In this embodiment, since the first signal line **151**, the second signal line **152**, and the third signal line **153** may be respectively electrically connected to the first conductive layer pattern **120**, each modulation unit **100** needs to be additionally provided with multiple radio frequency chokes **170** to block a high frequency signal or an alternating current voltage in the first conductive layer pattern **120** from entering (or interfering with) the first signal line **151**, the second signal line **152**, and the third signal line **153** with low frequency signals or direct current voltages. Specifically, the radio frequency chokes **170** are respectively disposed in the first signal line **151**, the second signal line **152**, and the third signal line **153** to stabilize the first voltage V**1**, the second voltage V**2**, and the third voltage V**3**.
(45) Although the first voltage V**1**, the second voltage V**2**, and the third voltage V**3** in this embodiment are all different from each other, the disclosure is not limited thereto. In other words, in some embodiments, the voltages of any two of the first voltage, the second voltage, and the third voltage may be different.
(46) Although the first signal line **151**, the second signal line **152**, and the third signal line **153** in this embodiment may be the same film layer as the first conductive layer pattern **120**, the disclosure is not limited thereto. In other words, in some embodiments, the first signal line, the second signal line, and the third signal line may also be different film layers from the first conductive layer pattern, as shown in FIG. **2**A, FIG. **2**B, FIG. **3**A, and FIG. **3**B. In some embodiments, one or two of the first signal line, the second signal line, and the third signal line may be the same film layer as the first conductive layer pattern, and the other one or the other two of the signal lines may be different film layers from the first conductive layer pattern, as shown in FIG. **6**.
(47) In this embodiment, in the schematic top view of the modulation unit **100** (as shown in FIG. **1**A), although the shapes of the contours of the first signal line **151**, the second signal line **152**, and the third signal line **153** are straight lines, the disclosure is not limited thereto. In some embodiments, the shapes of the contours of the first signal line, the second signal line, and the third signal line may also be wavy structures or loop structures.
(48) Other embodiments are listed below for illustration. It must be noted here that the following embodiments continue to use the reference numerals and some contents of the foregoing embodiment, wherein the same reference numeral is adopted to represent the same or similar elements, and the description of the same technical content is omitted. For the description of the omitted part, reference may be made to the foregoing embodiment, which will not be repeated in the following embodiments.
(49) FIG. **2**A is a schematic partial top view of an electronic device according to another embodiment. FIG. **2**B is a schematic cross-sectional view of the electronic device of FIG. **2**A along a section line II-II′. Please refer to FIG. **1**A, FIG. **1**B, FIG. **2**A, and FIG. **2**B at the same time. A

modulation unit **100***a* of this embodiment is substantially similar to the modulation unit **100** of FIG. **1**A to FIG. **1**B, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***a* of this embodiment is different from the modulation unit **100** in that each modulation unit **100***a* of this embodiment further includes a second insulation layer **132** and a second conductive layer pattern **180**.

(50) Specifically, please refer to FIG. **2**A and FIG. **2**B. In this embodiment, a section of another insulation layer, that is, the second insulation layer **132** is disposed on the first insulation layer **130** to cover at least one part of the first insulation layer **130** and expose another part of the first insulation layer **130**. The second insulation layer **132** may be a single-layer or multi-layer structure, and the material of the second insulation layer **132** may include a polymer thin film, silicon nitride, silicon oxide, silicon oxynitride, or a combination of the foregoing, but not limited thereto.

(51) In this embodiment, a first signal line **151***a*, a second signal line **152***a*, and a third signal line **153***a* are disposed on the first insulation layer **130** exposed by the second insulation layer **132**, so that the first signal line **151***a* is physically separated from the first part **121**, the second signal line **152***a* is physically separated from the second part **122**, and the third signal line **153***a* is physically separated from the third part **123**. The second conductive layer pattern **180** is disposed on the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a*. In other words, the second conductive layer pattern **180** may be disposed between a signal line **150** and an electronic element **140**, wherein the signal line **150** includes the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a*, and the electronic element **140** includes the first electronic element **141** and the second electronic element **142**. The second conductive layer pattern **180** includes a first pad **181**, a second pad **182**, and a third pad **183** that are separated from each other. The first pad **181** may be disposed corresponding to the first part **121**, the second pad **182** may be disposed corresponding to the second part **122**, and the third pad **183** may be disposed corresponding to the third part **123**.

(52) In this embodiment, the first signal line **151***a* may contact and be electrically connected to the first pad **181**, the second signal line **152***a* may contact and be electrically connected to the second pad **182**, and the third signal line **153***a* may contact and be electrically connected to the third pad **183**. The first pad **181** may be electrically connected to the first signal line **151***a* and the first electronic element **141**, the second pad **182** may be electrically connected to the second signal line **152***a* and the second electronic element **142**, and the third pad **183** may be electrically connected to the third signal line **153***a* and the first electronic element **141** and/or the second electronic element **142**.

(53) In this embodiment, since the high frequency signal or the alternating current voltage in the first conductive layer pattern **120** will not enter (or interfere with) the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a* with low frequency signals or direct current voltages, each modulation unit **100***a* does not need to be additionally provided with multiple radio frequency chokes.

(54) In this embodiment, since the first conductive layer pattern **120** is indirectly electrically connected to the first electronic element **141** and the second electronic element **142**, the first electronic element **141** and the second electronic element **142** need to regulate an electromagnetic wave signal (or a light signal, but not limited to) between the first conductive layer patterns **120** through induction.

(55) In this embodiment, a thickness T of the first insulation layer **130** may be 100 angstroms (Å) to 10 micrometers (μm), wherein the thickness T is the minimum thickness of the first insulation layer **130** measured along a normal direction of the substrate **101**.

(56) FIG. **3**A is a schematic partial top view of an electronic device according to another embodiment of the disclosure. FIG. **3**B is a schematic cross-sectional view of the electronic device of FIG. **3**A along a section line III-III′. Please refer to FIG. **2**A, FIG. **2**B, FIG. **3**A, and FIG. **3**B at the same time. A modulation unit **100***b* of this embodiment is substantially similar to the

modulation unit **100***a* of FIG. **2**A and FIG. **2**B, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***b* of this embodiment is different from the modulation unit **100***a* in that each modulation unit **100***b* in this embodiment further includes a conductive hole **190**.

(57) Specifically, please refer to FIG. **3**A and FIG. **3**B. In this embodiment, the conductive hole **190** may penetrate the first insulation layer **130** and electrically connect the second conductive layer pattern **180** and the first conductive layer pattern **120**. Since the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a* may be electrically connected to the first conductive layer pattern **120** through the second conductive layer pattern **180** and the conductive hole **190**, the modulation unit **100***b* of this embodiment needs to be additionally provided with multiple radio frequency chokes **170** in the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a*.

(58) In addition, since the first conductive layer pattern **120** may be electrically connected to the first electronic element **141** and the second electronic element **142** through the conductive hole **190** and the second conductive layer pattern **180**, the modulation unit **100***b* of this embodiment does not need to limit the thickness of the first insulation layer **130**.

(59) FIG. **4**A is a schematic partial top view of an electronic device according to another embodiment of the disclosure. FIG. **4**B is a schematic cross-sectional view of the electronic device of FIG. **4**A along a section line IV-IV′. Please refer to FIG. **2**A, FIG. **2**B, FIG. **4**A, and FIG. **4**B at the same time. A modulation unit **100***c* of this embodiment is substantially similar to the modulation unit **100***a* of FIG. **2**A and FIG. **2**B, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***c* of this embodiment is different from the modulation unit **100***a* in that the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a* of this embodiment do not contact the second conductive layer pattern **180**.

(60) Specifically, please refer to FIG. **4**A and FIG. **4**B. In this embodiment, the first insulation layer **130** has multiple openings to expose the first part **121**, the second part **122**, and the third part **123** of the first conductive layer pattern **120**.

(61) In this embodiment, the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a* are disposed on the first conductive layer pattern **120** exposed by the first insulation layer **130**, so that the first signal line **151***a* may contact the first part **121**, the second signal line **152***a* may contact the second part **122**, and the third signal line **153***a* may contact the third part **123**.

(62) In this embodiment, the second conductive layer pattern **180** is disposed on the first conductive layer pattern **120** exposed by the first insulation layer **130**, so that the first pad **181** may contact the first part **121**, the second pad **182** may contact the second part **122**, and the third pad **183** may contact the third part **123**. The first pad **181** does not contact the first signal line **151***a*, the second pad **182** does not contact the second signal line **152***a*, and the third pad **183** does not contact the third signal line **153***a*. The first pad **181** may be electrically connected to the first signal line **151***a* through the first part **121**, the second pad **182** may be electrically connected to the second signal line **152***a* through the second part **122**, and the third pad **183** may be electrically connected to the third signal line through the third part **123 153***a*.

(63) The second insulation layer **132** is disposed on the first insulation layer **130** to cover the first insulation layer **130**, the first signal line **151***a*, the second signal line **152***a*, and the third signal line **153***a*. The second insulation layer **132** may also be disposed in a gap between the first pad **181** and the first signal line **151***a*, a gap between the second pad **182** and the second signal line **152***a*, and a gap between the third pad **183** and the third signal line **153***a*.

(64) FIG. **5** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **5** at the same time. A modulation unit **100***d* of this embodiment is substantially similar to the modulation unit **100** of FIG. **1**A, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***d* of this embodiment is different from the modulation unit **100** in that in each modulation unit

100*d* of this embodiment, a pattern of a first conductive layer pattern **120***d* may be roughly regarded as a hollow square shape.

(65) Specifically, please refer to FIG. **5**. In the schematic top view of the modulation unit **100***d*, the pattern of the first part **121** and the pattern of the second part **122** may be regarded as L-shaped, and the pattern of the third part **123** may be regarded as C-shaped, but not limited thereto. The pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** are separated from each other, and the pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** may be roughly formed into a hollow square shape, but not limited thereto. In addition, in this embodiment, the first electronic element **141** and the second electronic element **142** are respectively disposed on two sides of the pattern of the first conductive layer pattern **120***d*, but not limited thereto.

(66) FIG. **6** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **2**A and FIG. **6** at the same time. A modulation unit **100***e* of this embodiment is substantially similar to the modulation unit **100** of FIG. **1**A, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***e* of this embodiment is different from the modulation unit **100** in that in each modulation unit **100***e* of this embodiment, a first electronic element **141***d* and a second electronic element **142***d* both have 3 pads (not shown).

(67) Specifically, please refer to FIG. **6**. In the schematic top view of the modulation unit **100***e*, the first conductive layer pattern **120***e* includes the first part **121**, the second part **122**, and the third part **123**. The pattern of the first part **121** and the pattern of the second part **122** may be regarded as a solid square shape, and the pattern of the third part **123** may be regarded as custom character-shaped, but not limited thereto. The pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** are separated from each other, and the pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** may be roughly formed into a custom character shape (that is, the pattern of the first conductive layer pattern **120***e* may be roughly regarded as the custom character shape), but not limited thereto. In addition, in this embodiment, the first electronic element **141***d* and the second electronic element **142***d* are respectively disposed on two sides of the pattern of the first conductive layer pattern **120***e*, but not limited thereto.

(68) In this embodiment, the second conductive layer pattern **180***d* is disposed on the first conductive layer pattern **120***e* and includes the first pad **181** and the second pad **182** separated from each other. The first pad **181** may be disposed corresponding to the first part **121**, and the second pad **182** may be disposed corresponding to the second part **122**. The bonding pad **161**, the bonding pad **162**, the bonding pad **163**, and the bonding pad **164** are respectively disposed on the third part **123** of the first conductive layer pattern **120***e* to contact the third part **123**. The first electronic element **141** having 3 pads may be bonded to the first conductive layer pattern **120***e* through the bonding pad **161**, the first pad **181**, and the bonding pad **163**, and the second electronic element **142** having 3 pads may be bonded to the first conductive layer pattern **120***e* through the bonding pad **162**, the second pad **182**, and the bonding pad **164**.

(69) In this embodiment, since the third signal line **153** may contact and be electrically connected to the third part **123** of the first conductive layer pattern **120***e*, the third signal line **153** needs to be additionally provided with the radio frequency choke **170**. In addition, the first signal line **151***d* electrically connected to the first pad **181** and the second signal line **152***d* electrically connected to the second pad **182** do not need to be additionally provided with any radio frequency choke.

(70) FIG. **7** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **7** at the same time. A modulation unit **100***f* of this embodiment is substantially similar to the modulation unit **100** of FIG. **1**A, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***f* of this embodiment is different from the modulation unit **100** in that each modulation unit **100***f*

of this embodiment further includes a fourth signal line **154**, and a first conductive layer pattern **120***f* further includes a fourth part **124**, a fifth part **125**, a fifth part **126**, a fifth part **127**, and a fifth part **128**.
(71) Specifically, please refer to FIG. **7**. In this embodiment, the first part **121**, the second part **122**, the third part **123**, the fourth part **124**, the fifth part **125**, the fifth part **126**, the fifth part **127**, and the fifth part **128** are separated from each other. The first part **121**, the second part **122**, the third part **123**, and the fourth part **124** are disposed in the middle of the first conductive layer pattern **120***f*, and the fifth part **125**, the fifth part **126**, the fifth part **127**, and the fifth part **128** are disposed at the periphery of the first conductive layer pattern **120***f*. The patterns of the first part **121**, the second part **122**, the third part **123**, the fourth part **124**, the fifth part **125**, the fifth part **126**, the fifth part **127**, and the fifth part **128** may be roughly formed into a custom character shape (that is, the pattern of the first conductive layer pattern **120***f* may be roughly regarded as the custom character shape), but not limited thereto.
(72) In this embodiment, the bonding pad **161** is disposed on the first part **121**, the bonding pad **162** is disposed on the second part **122**, the bonding pad **163** is disposed on the third part **123**, and the bonding pad **164** is disposed on the fourth part **124**. The first electronic element **141** may be bonded to the first conductive layer pattern **120***f* through the bonding pad **161** and the bonding pad **163**, and the second electronic element **142** may be bonded to the first conductive layer pattern **120***f* through the bonding pad **162** and the bonding pad **164**.
(73) In this embodiment, the first signal line **151** may contact and be electrically connected to the first part **121**, the second signal line **152** may contact and be electrically connected to the second part **122**, the third signal line **153** may contact and be electrically connected to the third part **123**, and the fourth signal line **154** may contact and be electrically connected to the fourth part **124**. The first signal line **151** may provide the first voltage V**1** to the first electronic element **141**, and the third signal line **153** may provide the third voltage V**3** to the first electronic element **141**. The second signal line **152** may provide the second voltage V**2** to the second electronic element **142**, and the fourth signal line **154** may provide a fourth voltage V**4** to the second electronic element **142**.
(74) FIG. **8** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **8** at the same time. An electronic device **10***g* and a modulation unit **100***g* of this embodiment are substantially similar to the electronic device **10** and the modulation unit **100** of FIG. **1**A and FIG. **1**B, so the same and similar components in the two embodiments are repeated here. The electronic device **10***g* and the modulation unit **100***g* of this embodiment are different from the electronic device **10** and the modulation unit **100** in that the electronic device **10***g* of this embodiment further includes multiple scanning lines SL, multiple signal lines DL, and a modulation module **11**, and each modulation unit **100***g* further includes a chip C.
(75) Specifically, please refer to FIG. **8**. The modulation module **11** includes multiple modulation units **100***g* arranged in an array, wherein the modulation units **100***g* disposed in the modulation module **11** may be the same or different. The scanning lines SL and the signal lines DL are respectively disposed on the substrate **101**. The scanning lines SL and the signal lines DL are interleaved with each other to define the modulation units **100***g* arranged in an array.
(76) In this embodiment, since the scanning lines SL may be electrically connected to the chip C, the signal lines DL may be electrically connected to the chip C, and the first part **121**, the second part **122**, and the third part **123** in the first conductive layer pattern **120** may be respectively electrically connected to the chip C through the first signal line **151**, the second signal line **152**, and the third signal line **153**, the chip C may independently regulate each modulation unit **100***g*. In this embodiment, the chip C may be, for example, a chip packaged by an integrated circuit (IC) or multiple thin film transistor (TFT) elements, or a die formed by multiple TFT elements. The chip C includes a driving circuit, and the driving circuit may be respectively electrically connected to the

first signal line **151**, the second signal line **152**, and the third signal line **153**.
(77) FIG. **9** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **8** and FIG. **9** at the same time. An electronic device **10***h* of this embodiment is substantially similar to the electronic device **10***g* of FIG. **8**, so the same and similar components in the two embodiments are not repeated here. The electronic device **10***h* in this embodiment is different from the electronic device **10***g* in that the electronic device **10***h* in this embodiment further includes an emission source **12**.
(78) Specifically, please refer to FIG. **9**. In this embodiment, the emission source **12** is configured to emit a signal EM (for example, an electromagnetic wave signal or a light signal). The signal EM (for example, the electromagnetic wave signal or the light signal) may propagate in, for example, a free space in any direction, but not limited thereto. In some embodiments, the signal EM (for example, the electromagnetic wave signal or the light signal) may also propagate in a wave-guided structure (not shown), wherein the wave-guided structure is, for example, a transmission line, a dielectric-filled waveguide, or a gas-filled waveguide, but not limited thereto.
(79) In this embodiment, when each modulation unit **100***h* in the modulation module **11** receives the signal EM (for example, the electromagnetic wave signal or the light signal) emitted by the emission source **12**, the modulation module **11** may regulate the phase, the amplitude, or the polarization state of the signal EM (for example, the electromagnetic wave signal or the light signal) incident to each modulation unit **100***h*, so that each modulation unit **100***h* outputs the signal EM (for example, the electromagnetic wave signal or the light signal) to the same direction (that is, a first direction D**1** or a second direction D**2**), but not limited thereto. In some embodiments, after the modulation module **11** regulates the phase of the signal EM (for example, the electromagnetic wave signal or the light signal) incident to each modulation unit **100***h*, each modulation unit **100***h* may also output the signal EM (for example, the electromagnetic wave signal or the light signal) to the same location (not shown).
(80) In this embodiment, the electronic device **10***h* may be applied to an antenna device, a display device for images, or a **5**G millimeter-wave amplifier, but not limited thereto.
(81) In addition, although the modulation module **11** in this embodiment is disposed outside the emission source **12** to regulate the phase of the signal EM (for example, the electromagnetic wave signal or the light signal) emitted by the emission source **12**, the disclosure does not limit the configuration location of the modulation module **11**. In some embodiments, a modulation module may be disposed in an emission source to regulate a phase of a signal (for example, an electromagnetic wave signal or a light signal) before emitting the signal, as shown in FIG. **10**.
(82) FIG. **10** is a functional schematic view of an electronic device according to another embodiment of the disclosure.
(83) Please refer to FIG. **10**. An electronic device **20** of this embodiment may be regarded as an antenna device and includes a baseband circuit **21** and a phase array antenna **22**. Specifically, the phase array antenna **22** includes multiple antennas **200**, and each antenna **200** includes a converter **220**, a converter **230**, an intermediate frequency circuit **240** and a high frequency circuit **260**. The intermediate frequency circuit **240** includes a filter **241**, an amplifier **242**, a mixer **243**, a local oscillator **244**, and an amplifier **245**. The high frequency circuit **260** includes a mixer **261**, a phase shifter **262**, a power amplifier **263**, a diplexer **264**, an antenna unit **265**, and a low noise amplifier **266**.
(84) In this embodiment, when the electronic device **20** emits a signal, the baseband circuit **21** first converts the signal to the intermediate frequency circuit **240** through the converter **220**, the signal is then transmitted to the high frequency circuit **260** through the filter **241**, the amplifier **242**, the mixer **243**, and the local oscillator **244** in the intermediate frequency circuit **240**, then through the mixer **261**, the phase shifter **262**, the power amplifier **263**, and the diplexer **264** in the high frequency circuit **260**, and an electromagnetic wave signal is finally emitted to the outside through the antenna unit **265**. On the contrary, when the electronic device **20** receives a signal, the antenna

unit **265** is first used to receive an electromagnetic wave signal from the outside, and the signal is then transmitted to the baseband circuit **21** sequentially through the high frequency circuit **260**, the intermediate frequency circuit **240**, and the converter **230**. A modulation module (not shown) or a modulation unit (not shown) may be disposed in the antenna unit **265** to regulate the phase, the bandwidth, the intensity, or the polarization state of the electromagnetic wave signal before emitting the electromagnetic wave signal.

(85) FIG. **11** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **11** at the same time. An electronic device **10***i* and a modulation unit **100***i* of this embodiment are substantially similar to the electronic device **10** and the modulation unit **100** of FIG. **1**A and FIG. **1**B, so the same and similar components in the two embodiments are not repeated here. The electronic device **10***i* and the modulation unit **100***i* of this embodiment are different from the electronic device **10** and the modulation unit **100** in that the electronic device **10***i* of this embodiment further includes multiple scanning lines SL, multiple signal lines DL, and a modulation module **11***i*, and each modulation unit **100***i* further includes multiple transistors TFT**1**, TFT**2**, and TFT**3**.

(86) Specifically, please refer to FIG. **11**. The modulation module **11***i* includes multiple modulation units **100***i* arranged in an array, wherein the modulation units **100***i* disposed in the modulation module **11***i* may be the same or different. The scanning lines SL and the signal lines DL are respectively disposed on the substrate **101**. The scanning lines SL and the signal lines DL are interleaved with each other to define the modulation units **100***i* arranged in an array.

(87) In this embodiment, the scanning lines SL may be electrically connected to gates G in the transistor TFT**1**, the transistor TFT**2**, and the transistor TFT**3**, the signal lines DL may be electrically connected to sources S in the transistor TFT**1**, the transistor TFT**2**, and the transistor TFT**3**, and the first conductive layer pattern **120** may be electrically connected to drains D in the transistor TFT**1**, the transistor TFT**2**, and the transistor TFT**3**. The transistor TFT**1**, the transistor TFT**2**, and the transistor TFT**3** may be respectively electrically connected to the first signal line **151**, the second signal line **152**, and the third signal line **153** to independently regulate each modulation unit **100***i*.

(88) FIG. **12** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **12** at the same time. An electronic device **10***j* and a modulation unit **100***j* of this embodiment are substantially similar to the electronic device **10** and the modulation unit **100** of FIG. **1**A and FIG. **1**B, so the same and similar components in the two embodiments are not repeated here. The electronic device **10***j* and the modulation unit **100***j* of this embodiment are different from the electronic device **10** and the modulation unit **100** in that the electronic device **10***j* of this embodiment further includes multiple scanning lines SL, multiple signal lines DL, a chip C, and a modulation module **11***j*.

(89) Specifically, please refer to FIG. **12**. The modulation module **11***j* includes multiple modulation units **100***j* arranged in an array, wherein the modulation units **100***j* disposed in the modulation module **11***j* may be the same or different. The scanning lines SL and the signal lines DL are respectively disposed on the substrate **101**. The scanning lines SL and the signal lines DL are interleaved with each other to define the modulation units **100***j*.

(90) In this embodiment, the chip C may be, for example, a chip packaged by an IC or multiple TFT elements, or a die formed by multiple TFT elements. The chip C may include a driving circuit. Since the chip C may be respectively electrically connected to the first part **121**, the second part **122**, and the third part **123** in each modulation unit **100***j* through the first signal line **151**, the second signal line **152**, and the third signal line **153**, the chip C may simultaneously drive multiple modulation units **100***j*.

(91) FIG. **13** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **13** at the same time. A modulation unit **100***k* of this embodiment is substantially similar to the modulation unit **100** of FIG. **1**A, so the

same and similar components in the two embodiments are not repeated here. The modulation unit **100***k* of this embodiment is different from the modulation unit **100** in that in each modulation unit **100***k* of this embodiment, a first conductive layer pattern **120***k* further includes a fourth part **124***k*, and a second electronic element **142***k* has 3 pads (not shown).

(92) Specifically, please refer to FIG. **13**. In this embodiment, the pattern of the first part **121**, the pattern of the second part **122**, the pattern of the third part **123**, and the pattern of the fourth part **124** are separated from each other, and the pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** may be roughly formed into a hollow square shape, but not limited thereto. The pattern of the first part **121** may be regarded as L-shaped, the pattern of the second part **122** may be regarded as a pattern composed of multiple separate solid square shapes, the pattern of the third part **123** may be regarded as a solid square shape, and the pattern of the fourth part **124** may be regarded as a solid square shape, but not limited thereto.

(93) In this embodiment, the bonding pad **161** and the bonding pad **162** are respectively disposed on the first part **121**, the bonding pad **163** is disposed on the second part **122**, the bonding pad **164** is disposed on the third part **123**, and the bonding pad **165** is disposed on the fourth part **124**. The first electronic element **141** may be bonded to the first conductive layer pattern **120***k* through the bonding pad **161** and the bonding pad **163**, and the second electronic element **142***k* having 3 pads may be bonded to the first conductive layer pattern **120***k* through the bonding pad **162**, the bonding pad **164**, and the bonding pad **165**.

(94) In this embodiment, the first signal line **151** may contact and be electrically connected to the first part **121**, the second signal line **152** may respectively contact and be electrically connected to the second part **122** and the third part **123**, and the third signal line **153** may contact and be electrically connected to the fourth part **124***k*. The first signal line **151** may provide the first voltage V**1** to the first electronic element **141**, and the second signal line **152** may provide the second voltage V**2** to the first electronic element **141**. The first signal line **151** may also provide the first voltage V**1** to the second electronic element **142***k*, the second signal line **152** may also provide the second voltage V**2** to the second electronic element **142***k*, and the third signal line **153** may provide the third voltage V**3** to the second electronic element **142***k*.

(95) FIG. **14** is a schematic partial top view of an electronic device according to another embodiment of the disclosure. Please refer to FIG. **1**A and FIG. **14** at the same time. A modulation unit **100***m* of this embodiment is substantially similar to the modulation unit **100** of FIG. **1**A, so the same and similar components in the two embodiments are not repeated here. The modulation unit **100***m* in this embodiment is different from the modulation unit **100** in that in each modulation unit **100***m* of this embodiment, a first conductive layer pattern **120***m* further includes an opening region **129**.

(96) Specifically, please refer to FIG. **14**. In this embodiment, the pattern of the first part **121**, the pattern of the second part **122**, and the pattern of the third part **123** are separated from each other, and the pattern of the first part **121** and the pattern of the third part **123** are disposed in the hollow of the pattern of the second part **122**, but not limited thereto. The pattern of the first part **121** may be regarded as a hollow semicircular shape, the pattern of the second part **122** may be regarded as a hollow square shape, and the pattern of the third part **123** may be regarded as a hollow semicircular shape, but not limited thereto.

(97) In this embodiment, the opening region **129** is a region exposed by the first part **121**, the second part **122**, and the third part **123** of the first conductive layer pattern **120***m*. The sum of the area of the first part **121**, the area of the second part **122**, and the area of the third part **123** is greater than the area of the opening region **129**.

(98) In this embodiment, the bonding pad **161** and the bonding pad **162** are respectively disposed on the second part **122**, the bonding pad **163** is disposed on the third part **123**, and the bonding pad **164** is disposed on the first part **121**. The first electronic element **141** may be bonded to the first conductive layer pattern **120***m* through the bonding pad **161** and the bonding pad **163**, and the

second electronic element **142** may be bonded to the first conductive layer pattern **120***m* through the bonding pad **162** and the bonding pad **164**.

(99) In this embodiment, the first signal line **151** may contact and be electrically connected to the first part **121**, the second signal line **152** may contact and be electrically connected to the second part **122**, and the third signal line **153** may contact and be electrically connected to the third part **123**. The second signal line **152** may provide the second voltage V**2** to the first electronic element **141**, and the third signal line **153** may provide the third voltage V**3** to the first electronic element **141**. The first signal line **151** may provide the first voltage V**1** to the second electronic element **142**, and the second signal line **152** may also provide the second voltage V**2** to the second electronic element **142**.

(100) FIG. **15** to FIG. **17** are schematic partial top views of an electronic device according to multiple embodiments of the disclosure. Please refer to FIG. **1**A and FIG. **15** to FIG. **17** at the same time. Modulation units **100***n*, **100***p*, and **100***q* of this embodiment are substantially similar to the modulation unit **100** of FIG. **1**A, so the same and similar components in the two embodiments are not repeated here. The differences between the modulation units **100***n*, **100***p*, and **100***q* of this embodiment and the modulation unit **100** are as follows.

(101) Please refer to FIG. **15**. In the modulation unit **100***n* of this embodiment, the shapes of the contours of a first signal line **151***n*, a second signal line **152***n*, and a third signal line **153***n* may be arc-shaped wavy structures, so that the first signal line **151***n*, the second signal line **152***n*, and the third signal line **153***n* do not need to be additionally provided with radio frequency chokes.

(102) Please refer to FIG. **16**. In the modulation unit **100***p* of this embodiment, the shapes of the contours of a first signal line **151***p*, a second signal line **152***p*, and a third signal line **153***p* may be rectangular wavy structures, so that the first signal line **151***p*, the second signal line **152***p*, and the third signal line **153***p* do not need to be additionally provided with radio frequency chokes.

(103) Please refer to FIG. **17**. In the modulation unit **100***q* of this embodiment, the shapes of the contours of a first signal line **151***q*, a second signal line **152***q*, and a third signal line **153***q* may be loop structures, so that the first signal line **151***q*, the second signal line **152***q*, and the third signal line **153***q* do not need to be additionally provided with radio frequency chokes.

(104) In summary, in the electronic device of the embodiments of the disclosure, since the first voltage and the third voltage may regulate the own characteristics of the first electronic element, and the second voltage and the third voltage may regulate the own characteristics of the second electronic element, the modulation unit may modulate the intensity, the bandwidth, or the phase of the received signal (for example, the electromagnetic wave signal or the light signal). Then, since the first electronic element and the second electronic element may share the third voltage of the third signal line, the overall circuit configuration of the modulation unit may be relatively simple and not complicated. Furthermore, since the first voltage may be different from the second voltage, and the third voltage may be different from the first voltage and the second voltage, the regulated first electronic element and second electronic element may respectively display various different characteristics, thereby increasing the modulable factor of the modulation unit or increasing the selection of the first electronic element and the second electronic element.

(105) Finally, it should be noted that the above embodiments are only used to illustrate, but not to limit, the technical solutions of the disclosure. Although the disclosure has been described in detail with reference to the above embodiments, persons skilled in the art should understand that the technical solutions described in the above embodiments can still be modified or some or all of the technical features thereof can be equivalently replaced. However, the modifications or replacements do not cause the essence of the corresponding technical solutions to deviate from the scope of the technical solutions of the embodiments of the disclosure.

Claims

1. An electronic device, comprising: a substrate; and a plurality of modulation units, disposed on the substrate, wherein each of the modulation units comprises: a first electronic element and a second electronic element; a first signal line, providing a first voltage to the first electronic element; a second signal line, providing a second voltage to the second electronic element; a third signal line, providing a third voltage to the first electronic element and/or the second electronic element; a first conductive layer pattern, disposed on the substrate; and a first insulation layer, disposed on the first conductive layer pattern, wherein the first voltage is different from the second voltage, and the third voltage is different from the first voltage and/or the second voltage, wherein the first signal line, the second signal line, and the third signal line are disposed on the first insulation layer, wherein the first electronic element and the second electronic element are disposed on the first signal line, the second signal line, and the third signal line.
2. The electronic device according to claim 1, wherein each of the modulation units further comprises: a plurality of radio frequency chokes, respectively disposed in the first signal line, the second signal line, and the third signal line to stabilize the first voltage, the second voltage, and the third voltage.
3. The electronic device according to claim 1, wherein the first conductive layer pattern comprises a first part, a second part, and a third part separated from each other, wherein the first signal line is coupled to the first part, the second signal line is coupled to the second part, and the third signal line is coupled to the third part.
4. The electronic device according to claim 3, wherein the first signal line is physically separated from the first part, the second signal line is physically separated from the second part, and the third signal line is physically separated from the third part.
5. The electronic device according to claim 1, wherein each of the modulation units further comprises: a second conductive layer pattern, disposed between a signal line and an electronic element, and comprising a first pad, a second pad, and a third pad separated from each other, wherein the first pad is electrically connected to the first signal line and the first electronic element, the second pad is electrically connected to the second signal line and the second electronic element, and the third pad is electrically connected to the third signal line and the first electronic element and/or the second electronic element.
6. The electronic device according to claim 5, wherein the first signal line contacts the first pad, the second signal line contacts the second pad, and the third signal line contacts the third pad.
7. The electronic device according to claim 5, wherein each of the modulation units further comprises: a conductive hole, penetrating the first insulation layer and electrically connecting the second conductive layer pattern and the first conductive layer pattern.
8. The electronic device according to claim 1, wherein the first signal line, the second signal line, and the third signal line are disposed on the first conductive layer pattern.
9. The electronic device according to claim 8, wherein the first conductive layer pattern comprises a first part, a second part, and a third part separated from each other, wherein the first signal line contacts the first part, the second signal line contacts the second part, and the third signal line contacts the third part.
10. The electronic device according to claim 9, wherein each of the modulation units further comprises: a second conductive layer pattern, disposed between a signal line and an electronic element, and comprising a first pad, a second pad, and a third pad separated from each other, wherein the first pad is electrically connected to the first signal line and the first electronic element, the second pad is electrically connected to the second signal line and the second electronic element, and the third pad is electrically connected to the third signal line and the first electronic element and/or the second electronic element.
11. The electronic device according to claim 10, wherein the first pad contacts the first part, the second pad contacts the second part, and the third pad contacts the third part.

12. The electronic device according to claim 1, wherein each of the modulation units further comprises: a plurality of transistors, respectively electrically connected to the first signal line, the second signal line, and the third signal line.
13. The electronic device according to claim 1, wherein each of the modulation units further comprises: a driving circuit, respectively electrically connected to the first signal line, the second signal line, and the third signal line.
14. The electronic device according to claim 1, further comprising: an emission source, configured to emit a signal, wherein each of the modulation units receives the signal, and modulates a phase, an amplitude, or a polarization state of the signal.
15. The electronic device according to claim 1, wherein the first electronic element and the second electronic element comprise capacitors, inductors, or resistors.